# When should it switch?

## IoT · Level 3 · Lesson 2

# Today you will

- Choose a threshold for a rule from real readings.
- Explain false alarms and misses, and decide which costs more.
- Use two thresholds so a device doesn't flick on and off.

# Choosing the number

- Choose from real readings
- False alarm or miss: which costs more?
- Too many false alarms: nobody listens
- Two thresholds stop the flicker

# Find the two thresholds

1. Choose a device that switches itself: a fridge or a heater.
2. Make a table: time on, time off.
3. Listen for an hour and fill in the table.
4. Work out how long it runs and how long it rests.
5. Decide: one threshold, or two?

**Don't open any device, touch wiring or hot parts, or unplug anything: just listen and watch. Ask before you go into someone else's room.**

# Let’s discuss

1. Which alarm in your life cries wolf, and what number would you change?
2. When is a miss worse than a false alarm?
3. Why does a fridge run for minutes, not seconds?

# Check yourself

- Give one alarm where a miss costs more, and one where a false alarm costs more.